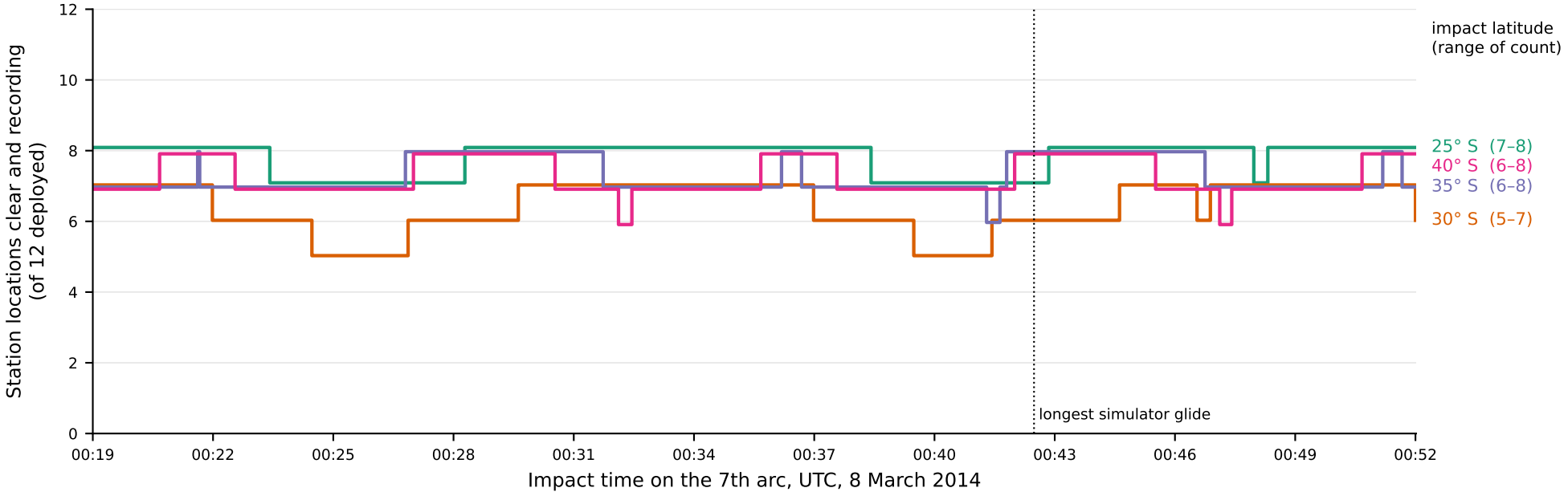
For an impact on the 7th arc at 25° S, 30° S, 35° S or 40° S between 00:19 and 00:52 UTC, 5 to 8 station locations had clear line of sight and were recording when the sound would arrive



Plain-language note. Each line counts the stations that have a clear path from the impact point and were recording at the time the sound would arrive there. Stations at one place are counted once. The steps come from the Australian seabed loggers, which record 307 s in each 15 min. Lines are moved up or down by less than 0.1 so that equal counts stay visible; read the count from the nearest whole number. 25° S: counted throughout Cape Leeuwin, Diego Garcia South, SWAMS1-3, SSEIR, MAD, NCRO, WKER; part of the time Perth Canyon. 30° S: counted throughout Cape Leeuwin, Diego Garcia South, SWAMS1-3, MAD, WKER; part of the time Perth Canyon, Portland. 35° S: counted throughout Cape Leeuwin, SWAMS1-3, SSEIR, MAD, NCRO, WKER; part of the time Perth Canyon, Portland. 40° S: counted throughout Cape Leeuwin, Diego Garcia South, SWAMS1-3, SSEIR, NCRO, WKER; part of the time Perth Canyon, Portland. This does not show that the sound was loud enough to detect; that is the next step.

Technical note. Count of locations in state 'clear, recording' in the impact-time chart (arc_recording_opportunity.py; series in arc_recording_opportunity_clear_count_by_impact_time.csv). Ridge (partial loss) and large-attenuation paths are not counted. Line of sight: hydro_data.sqlite table arc_path (arc_line_of_sight.py; WGS84 geodesics on GEBCO 2026, 15 arc-s). Blocked: land or water shallower than min(receiver depth, 200 m) more than 20 km from the receiver. Large attenuation: >= 10 km shallower than 1,000 m more than 100 km from the receiver. Ridge: >= 10 km shallower than the hydrophone (deep receivers). These thresholds are agent choices, not ruled. Recording: IMOS record windows from table segment (UTC, clock-corrected, event-day and background sets joined; no record missing 00:00-05:00 UTC); OHASISBIO continuous at 240 Hz (Royer 2016), NEAMS lost during recovery; IMS H01W/H08S continuous 250 Hz (8 Mar 2014 data used in Kadri 2024, CMST 2014); H08N continuous assumed (no published use of its 8 Mar 2014 data found). A location takes its most favourable instrument that is recording. Travel time = geodesic distance / c, c = 1.482 ± 0.006 km/s (registry constant c_sofar); 1-sigma timing error 4-16 s. Not included: HA04 Crozet (installed Dec 2016), IMOS deployments with no 8 Mar 2014 files, island seismometers. Geometric screen, not transmission loss; noise (for example the seismic survey shots Royer reports) is not included. Impact time axis 00:19:00-00:52:00 UTC at 1 s steps (agent choice, run J 99.9 % impact-time quantile after 00:19); dotted line 00:42:29 = longest Boeing-simulator glide. Arrival treated as an instant.